MIZORAM PUBLIC SERVICE COMMISSION

TECHNICAL COMPETITIVE EXAMINATION FOR JUNIOR GRADE OF MIZORAM HEALTH SERVICES (COMBINED) (DENTAL SURGEON SUB-CADRE) UNDER HEALTH & FAMILY WELFARE DEPARTMENT, GOVERNMENT OF MIZORAM. (SEPTEMBER-2025)

TECHNICAL SUBJECT PAPER-I (DENTAL SURGEON)

Time Allowed : 3 hours FM : 200

All questions carry equal marks of 2 each.
Attempt all questions.

1. Which tooth has the longest root?
(a) Maxillary central incisor (b) Maxillary canine
(c) Mandibular canine (d) Mandibular molar

2. Which dental structure is responsible for tooth sensitivity?
(a) Enamel (b) Cementum
(c) Dentin (d) Pulp

3. What is the function of the cusps on a molar tooth?
(a) To tear food (b) To grind food
(c) To crush food (d) To pierce food

4. While removing mandibular 3rd molar which part of bone should be used as a fulcrum?
(a) Lingual cortical bone (b) Mesial inter-radicular bone
(c) Buccal alveolar bone (d) Distal surface of adjacent crown

5. The chisel should be used with:
(a) Bevel towards the bone which is to be saved
(b) Bevel towards the bone which is to be sacrificed
(c) Bevel direction is not important
(d) Flat surface parallel to direction of grains of bone

6. While making vertical incision in Ward's incision one can injure :
(a) Buccal fat pad (b) Branches of lingual nerve
(c) Branches of facial nerve (d) Branches of facial artery/vein

7. The disadvantage of lingual split technique is/are:
(a) Injury to the lingual nerve
(b) Opening up of facial spaces on lingual side and floor of mouth
(c) Chances of dislodging tooth or root in sub-lingual space
(d) All of the above

8. The complication of using air rotor at 30,000rpm for impacted teeth removal is:
(a) Bone necrosis (b) Dehiscence
(c) Tissue laceration (d) Emphysema

9. In the extraction of mandibular third molars, the posterior incision is placed more:
(a) Bucally (b) Should not be placed
(c) Lingually (d) None of the above

10. Of the following the most difficult impaction to extract is:
(a) Mesioangular (b) Distoangular
(c) Vertical (d) Horizontal

11. The best time of extraction in pregnancy is:
(a) First trimester (b) Third trimester
(c) Second trimester (d) Contraindicated

12. The most important suture while closing Ward's incision is:
(a) Suturing of vertical limb
(b) Suturing of retromolar limb
(c) Suturing of area immediately distal to 2nd molar
(d) None of the above

13. A patient with class II Div I malocclusion undergoes genioplasty. His anterior teeth after the operation would be:
(a) Edge to edge bite (b) Without any change
(c) Having overjet of 2mm (d) No overbite

14. Sagittal split osteotomy is a procedure carried out for:
(a) Mandibular deformities (b) Tmj ankylosis
(c) Maxillary deformities (d) All of the above

15. Apertognathia is a condition in which there is:
(a) Retrogenia (b) Open bite deformity
(c) Maxillary hypoplasia (d) None of the above

16. Marsupialization is a surgical procedure which is commonly used to treat:
(a) Cystic odontoma (b) Large Cyst
(c) Periapical cyst (d) Oro antral fistula

17. Paraesthesia is a common finding in which of the following fractures?
(a) Subcondylar fracture (b) Zygomatic maxillary complex fracture
(c) Symphysis fracture (d) Coronoid fracture

18. A Gunning's splint is used in the treatment of:
(a) Anterior dentoalveolar fracture (b) Fracture of edentulous mandible
(c) Condylar neck fracture (d) All of the above

19. Guerin's fracture is synonymous with:
(a) Le fort I fracture (b) Le fort III fracture
(c) Le fort II fracture (d) Le fort IV fracture

20. Ecchymosis in the post auricular region over the mastoid process is called as:
(a) Guerin's sign (b) Murphy's sign
(c) Battle's sign (d) None of the above

21. Submentovertex view is an ideal view for diagnosing fracture of:
(a) NOE region
(b) Angle of mandible
(c) Zygomatic arch
(d) Condylar fracture

22. Osteomyelitis is more common in:
(a) Maxilla
(b) Mandible
(c) Zygoma
(d) Nasal complex

23. If OAF has been formed upon tooth extraction, the treatment would be:
(a) Immediate primary closure
(b) Sinus is irrigated, lavaged for 2/3 days
(c) Closure after 7 days
(d) None of the above

24. Cementum is thickest at:
(a) Apical third
(b) Furcation area
(c) Coronal third
(d) Middle third of root

25. Principal fibres that are absent in intermediate plexus:
(a) Oblique
(b) Transeptal
(c) Horizontal
(d) All of the above

26. Bacteria found in gingivitis are localized in:
(a) Connective tissue fibres
(b) Alveolar bone
(c) Gingival sulcus
(d) Periodontal ligament

27. Common clinical sign of occlusal trauma:
(a) Tooth mobility
(b) Resorption of alveolar ridge
(c) Fractures of cusps
(d) Widening of PD ligament

28. Crater like gingival deformities are seen in:
(a) ANUG
(b) Chronic gingivitis
(c) J.P
(d) Adult periodontitis

29. First clinical sign of juvenile periodontitis is:
(a) Pathological tooth migration and midline diastema
(b) Pain and bleeding
(c) Tooth mobility and bone loss
(d) Gingival enlargement and pus formation

30. Mrs. Rothuami , a 50 year old lady, has multiple periodontal abscess. The lab test you would do
(a) Biopsy to rule out carcinoma
(b) Blood analysis for diabetes
(c) Wassermen test
(d) Prothrombin test

31. The first antibody to be formed after encounter with an antigen is:
(a) IgM
(b) IgD
(c) IgA
(d) IgG

32. Sub gingival scaling and root planning is better carried by:
(a) Curet
(b) Universal scaler
(c) Sickle
(d) Ultrasonic scaler

33. A patient returned one week after scaling and prophylaxis. Hard black deposits of calculus are noted in gingival margin. This indicates:
(a) Patient is a heavy tea drinker
(b) Shrinkage of tissue due to reduction in inflammation after scaling
(c) Poor maintenance and formation of new calculus
(d) Blood clots which are not removed from tooth surface

34. Appropriate mode of therapy of gingival hyperplasia by dilantin sodium is:
(a) Flap surgery
(b) Gingivectomy and gingivoplasty
(c) Curettage
(d) Root planning

35. Graft taken from different individual of same species is:
(a) Allograft
(b) Xenograft
(c) Alloplast
(d) Autograft

36. The minimum concentration of chlorhexidine required to be effective as plaque inhibitor is:
(a) 0.22%
(b) 0.20%
(c) 0.12%
(d) 0.10%

37. Wickham's striae are seen in:
(a) Leukoplakia
(b) Erythroplakia
(c) Lichen planus
(d) Carcinoma in situ

38. Recurrent oral ulceration, atrophic glossitis, burning tongue, candidosis and angular cheilitis are suggestive of:
(a) Leukemia
(b) Thrombocytopenia
(c) Anemia
(d) All of the above

39. Butterfly rash on nose and malar prominence, dryness of mouth, inflamed areas with tendency to ulcerate in an adult female is most likely:
(a) Sun burn
(b) Lupus erythematosis
(c) Allergic reaction
(d) All of the above

40. Bence Jones protein in urine is a feature seen in:
(a) Diabetes mellitus
(b) Prostatic carcinoma
(c) Diabetes insipidus
(d) Multiple myeloma

41. The most common malignancy of the oral cavity is:
(a) Basal cell carcinoma
(b) Squamous cell carcinoma
(c) Fibrosarcoma
(d) Adenosarcoma

42. A white patch that cannot be rubbed off from oral mucousa is a feature of:
(a) Lichen planus
(b) Leukoplakia
(c) Erythroplakia
(d) Carcinoma

43. Tissue removal for microscopic examination should be immersed immendiately in:
(a) 10% alcohol
(b) 10% formaldehyde
(c) 10% formalin
(d) 5% ringers lactate solution

44. Cyst that has a highest recurrence rate is:
(a) Incisive canal cyst
(b) Odontogenic keratocyst
(c) Nasolabial cyst
(d) Radicular cyst

45. Which lesion is characterized by burning sensation preceded by vesicle formation, always with juxta epithelial inflammation (there is fibroelastic change in connective tissue leading to stiffness and trimus)?
(a) Leukoplakia
(b) Carcinoma in situ
(c) Erythroplakia
(d) Oral submucous fibrosis

46. A condition of the mouth which increases the caries activity in the oral cavity is:
(a) Xerostomia
(b) Thin saliva
(c) Malignancy
(d) Hairy tongue

47. *'Branchless fruit laden tree'* or *'cherry blossom appearance'* is radiographically seen in:
(a) Clear cell carcinoma
(b) Adeno cystic cell carcinoma
(c) Mucoepidermoid carcinoma
(d) Sjogren's syndrome

48. Radiation caries is a rampant form of dental decay primarily caused by irradiation of:
(a) Enamel, dentin and cementum
(b) Cervical fibres of periodontal ligament
(c) Pulps of teeth
(d) Salivary glands

49. The purpose of Aluminium filters in the path of X-ray beam is to:
(a) Increase contrast
(b) Reduce patient exposure
(c) Reduce film density
(d) Reduce film details

50. Developing solution is:
(a) Alkaline in nature
(b) Neutral in nature
(c) Acidic in nature
(d) None of the above

51. Brown discolouration seen in old radiographic film is due to:
(a) Over development
(b) Incomplete fixation
(c) Under development
(d) Exposure to light

52. What is the primary benefit of using cone beam computed tomography(CBCT) in dental radiology?
(a) Higher radiation dose
(b) Detailed 3D imaging
(c) Lower radiation dose
(d) Faster image acquisition

53. Which technology is used to capture digital impressions of teeth and soft tissues?
(a) Digital radiography
(b) Intraoral scanners
(c) Intraoral cameras
(d) CBCT

54. What is the benefit of using 3D printing in dental radiology?
(a) Creating accurate models for treatment planning
(b) Reducing radiation exposure
(c) Improving image quality
(d) Increasing patient comfort

55. What is the primary purpose of a pulp vitality test?
(a) To assess tooth mobility
(b) To evaluate periodontal health
(c) To determine the vitality of the dental pulp
(d) To evaluate tooth morphology

56. The apical end of a root canal filling in radiograph should be:
(a) Flush with the radiographic apex
(b) 1 to 1.5mm short of radiographic apex
(c) 3mm short of the radiographic apex
(d) 1mm extended beyond the radiographic apex

57. Condylar neck fractures can be best seen on:
(a) Reverse Towne's view
(b) Lateral view of skull
(c) PA view
(d) Water's view

58. The fastest dental film currently available has the speed rating of:
(a) 'D' speed
(b) 'F' speed
(c) 'E' speed
(d) 'C' speed

59. Glasgow coma scale is used:
(a) To ascertain motor response
(b) Eye response
(c) Verbal responsiveness
(d) To ascertain level of consciousness

60. Mrs. Parmawii, a patient complaint of sudden electric shock like pain, unilateral , short in duration but occurs again and again. Probable diagnosis would be:
(a) Neurosarcoma
(b) Cemento epithelial ossifying tumour
(c) Third molar impaction
(d) Trigeminal neuralgia

61. Which of the following teeth is most commonly associated with OAF?
(a) Maxillary first molar (b) Maxillary first premolar
(c) Maxillary second molar (d) Maxillary cuspid

62. Largest cusp in permanent mandibular first molar is:
(a) Mesiobuccal (b) Mesiolingual
(c) Distobuccal (d) Distal

63. What is the functional form of anterior teeth from the mesial or the distal aspect?
(a) Rhomboidal (b) Elliptical
(c) Trapezoidal (d) Wedge shaped

64. Steepest cusp is seen in:
(a) Maxillary 1st premolar (b) Maxillary 1st molar
(c) Mandibular 2nd molar (d) Mandibular 1st molar

65. The last primary tooth to be replaced by a permanent tooth is usually the:
(a) Maxillary second molar (b) Mandibular second molar
(c) Maxillary canine (d) Mandibular canine

66. Which of the following is dental formula of a 9 year old child:
(a) 6EDC21/6EDC21 (b) 6EDCBA/6EDC21
(c) EDC21/EDC21 (d) 76EDC21/76EDC21

67. Over jet is defined as:
(a) Horizontal overlap (b) Transverse plane discrepancy
(c) Vertical overlap (d) All of the above

68. The key to occlusion is :
(a) Maxillary permanent first molar (b) Maxillary permanent second molar
(c) Maxillary primary first molar (d) Maxillary primary second molar

69. Desmodont is another name for :
(a) The tooth with one wall pocket (b) Tooth with three walled pocket
(c) Periodontal ligament (d) Dehiscence

70. Basically TMJ is as :
(a) Ball and socket joint (b) Hinge and axis joint
(c) Synovial joint (d) Diarthrodal joint

71. Who among the following has the greatest risk of oral cancer?
(a) Chronic alcoholics, with a poor oro- dental hygiene and nutritional deficiencies
(b) Middle aged with poor oro-dental hygiene and regularly using tobacco
(c) Old, chronic alcoholics, regularly using tobacco
(d) Young with poor oro-dental hygiene and heavy smoking habit

72. The most common route for metastasis of oral cancer is by:
(a) Direct extension (b) Lymphatics
(c) Blood vessels (d) Aspiration of tumor cells

73. Which of the following is appropriate treatment modality for verrucous carcinoma?
(a) Surgery + Radiotherapy (b) Excision
(c) Laser technique (d) Alpha interferon

74. Which of the following white lesions disappears, on stretching the buccal mucosa ?
(a) Leukoplakia (b) Focal hyperkeratosis
(c) Linea alba (d) Leukoedema

75. Starry sky appearance is seen in:
(a) Pagets disease
(b) Cherubism
(c) Garrey's osteomyelitis
(d) Burkitts Lymphoma

76. Cancer which most commonly metastasizes to jaw bone is :
(a) Breast
(b) Prostatic
(c) Lung
(d) Kidney

77. Tobacco usage has been associated with:
(a) Hyperkeratosis
(b) Erythema
(c) Ulceration
(d) None of the above

78. Port wine stain is characteristic feature of:
(a) Peutz Jegers Syndrome
(b) Sturge weber Syndrome
(c) Albright's Syndrome
(d) Lymphangioma

79. X –rays are produced in :
(a) Anode
(b) Cathode
(c) Glass wall
(d) Molybdenum focusing cup

80. Units of radiation exposure is :
(a) Rad
(b) Roentgen
(c) Rem
(d) Seivert

81. Filters of metal that is used to remove long wave length x-rays from the primary beam is :
(a) Platinum
(b) Aluminium
(c) Copper
(d) Lead

82. Effect of radiation is :
(a) Cumulative
(b) Fractionalized
(c) Concentrated
(d) Dependent on genetic make up of individuals

83. Radiation induced caries occurs in?
(a) 1 months
(b) 3 months
(c) 6 months
(d) 9 months

84. Which of the following is the best method for radiation protection of the operator?
(a) Standing behind a lead barrier
(b) Wearing a lead apron
(c) Following the position and distance rule
(d) Standing 6 feet away from the x-ray tube during exposure

85. Interproximal caries is best detected by:
(a) Bitewing radiography
(b) Periapical
(c) Occlusal
(d) None of the above

86. In determining blood pressure by auscultatory method:
(a) The loudest sound is the diastolic pressure
(b) Systolic pressure estimation tends to be lower than those make by palpatory method
(c) The first sound heard is the systolic pressure
(d) The sounds that are heard are generated in heart

87. Tooth mobility is NOT reversible when there is :
(a) Inflammation in periodontium
(b) Trauma from occlusion
(c) Loss of alveolar bone
(d) Faulty tooth brushing technique

88. Infra-bony pockets are treated by :
(a) Gingivectomy (b) Gingivoplasty
(c) Flap operation (d) Vestibuloplasty

89. Which tooth brushing technique should be used in a patient with severe gingival recession and loss of gingival contour :
(a) Bass (b) Stillman
(c) Charter (d) Multitufted

90. Scaling and root planning is performed :
(a) As initial treatment (b) Post surgically
(c) In only gingival enlargements (d) After only antibiotics

91. Ultrasonic scalers vibrate at approximately
(a) 25000 cycles per second (b) 15000 cycles per second
(c) 10000 cycles per second (d) 50000 cycles per second

92. Mobility of teeth is measured using :
(a) Periodontometer (b) Periotron
(c) Spechtometer (d) Densitometer

93. Metronidazole is the drug of choice for :
(a) All periodontal disease (b) ANUG
(c) Juvenile periodontitis (d) Herpes simplex

94. Extraction of disto-angular impaction of mandibular 3rd molar can cause:
(a) Slippage in lingual pouch (b) Fracture of ramus of mandible
(c) Excessive Haemorrhage (d) Dry socket

95. A 23 year old female patient complains of decreased mouth opening since 4 days .This could be due to:
(a) Impacted third molar (b) Oral submucous fibrosis
(c) Oropharyngeal fibrosis (d) Bony ankylosis of TMJ

96. Which nerve is near to mandibular third molar (mesial aspect) and is likely to be damaged while removing 3rd molar ?
(a) Inferior alveolar nerve (b) Buccal nerve
(c) Massetric nerve (d) Lingual nerve

97. The longest acting LA is:
(a) Prilocaine (b) Ligocaine
(c) Bupivacaine (d) Ropivacaine

98. Number 16 cowhorn forceps are specially designed to extract :
(a) Maxillary molars (b) Mandibular molars
(c) Mandibular central incisors (d) Maxillary first and second premolars

99. Which surgical procedure is carried out to treat TMJ ankylosis?
(a) Arthroplasty (b) Condylectomy
(c) Discoplasty (d) Capsulorrhaphy

100. A pregnant patient in second trimester falls into syncope during extraction of upper molars. She should be kept in :
(a) Trendelenburg position (b) Head down towards her feet
(c) Right lateral position (d) Left lateral position

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